

■ Equalizer (10-Band EQ)

Feature Description

A 10-band graphic equalizer built with the Web Audio API. It inserts ten peaking BiquadFilter nodes (32 Hz – 16 kHz) between the <audio> element and the output, letting the user boost or cut each frequency band by ± 12 dB. Ten genre/mood presets (Flat, Bass Boost, Treble Boost, Vocal, Rock, Pop, Jazz, Electronic, Classical, Podcast) are included, plus a power toggle and a reset-to-flat button. Settings persist in localStorage across sessions. The EQ button appears in both the bottom player bar and the fullscreen player, and can be opened with the **E** keyboard shortcut.

Why It Was Added

Different music genres, headphones, and room acoustics benefit from frequency shaping. An equalizer lets listeners tailor the sound to their preference — boosting bass for workout playlists, lifting vocals for podcasts, or adding sparkle to acoustic tracks — without needing external system-level EQ software. It makes Rainy a more complete, self-contained listening experience.

Step-by-Step Testing Instructions

Step 1. Open Rainy in a browser and log in. Start playing any song.

Expected: Audio plays normally through the default output.

Step 2. Click the **Equalizer button** (slider icon) in the bottom player bar, to the right of the Playback Speed button.

Expected: A popup panel titled 'Equalizer' appears above the button, showing 10 vertical sliders (32 Hz to 16k Hz), a row of preset buttons, a power toggle (■), and a 'Reset to Flat' button.

Step 3. Click the **'Bass Boost'** preset button.

Expected: The first 3–4 sliders (32–250 Hz) jump upward to +6/+5/+4/+2 dB. The bass in the playing song becomes noticeably stronger. The 'Bass Boost' button is highlighted.

Step 4. Click the **'Treble Boost'** preset.

Expected: The high-frequency sliders (1k–16k Hz) move up; the sound becomes brighter/crisper. 'Treble Boost' is now highlighted and 'Bass Boost' is deselected.

Step 5. Manually drag the **1000 Hz slider** down to –6 dB.

Expected: The value label above the slider shows '–6'. All preset buttons deselect (custom curve). The midrange/vocal presence in the song is reduced.

Step 6. Click the **power toggle (■)** to turn the EQ off.

Expected: All sliders snap to 0 dB and the audio returns to flat/unprocessed sound. The power button loses its accent highlight.

Step 7. Click the power toggle again to **re-enable** the EQ.

Expected: The previously saved custom curve (including the -6 dB at 1 kHz) is restored and applied.

Step 8. Click '**Reset to Flat**'.

Expected: All sliders return to 0 dB, 'Flat' preset is highlighted, and audio is unprocessed.

Step 9. Close the EQ menu (click outside it). **Reload the page** and start a song. Open the EQ menu again.

Expected: The last-used gains and enabled/disabled state are restored from localStorage — settings survived the reload.

Step 10. Open the **fullscreen player** (press F or click the expand button) and click the EQ button there.

Expected: The same EQ popup appears anchored to the fullscreen EQ button and controls the same audio graph.

Step 11. Press the **E** key on the keyboard (while not typing in a field).

Expected: The EQ popup opens anchored to the bottom-bar EQ button.

Step 12. With the EQ open, press **Escape** or click outside the menu.

Expected: The EQ menu closes.

Edge Cases to Check

- Open the EQ before any song has started playing — the AudioContext should still be created on the button click (user gesture) without errors.
- Rapidly switch between presets while a song plays — audio should not glitch, stutter, or produce clicks.
- Set all sliders to $+12$ dB (maximum boost) — audio should not clip or distort badly (some loudness increase is expected).
- Set all sliders to -12 dB — audio should become very quiet but not silent or broken.
- Open the EQ menu from the bottom bar, then switch to fullscreen and open it there — only one menu should exist at a time (the first is removed).
- Use the EQ while the Sleep Timer or Playback Speed is also active — all three features should coexist without conflict.
- Clear localStorage (DevTools → Application → Local Storage → Clear All), reload, and open the EQ — it should default to Flat / disabled gracefully.
- Test in a browser that does not support AudioContext (rare) — a toast error 'Equalizer not supported' should appear and the app should not crash.
- Seek to a different position in the song while the EQ is active — the EQ processing should continue seamlessly.
- Switch to a different song (next/prev) while the EQ is active — the EQ should remain applied to the new track without needing to re-open the menu.